Equity Foundations

Trading Venues and Stylized Facts

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1 Market Microstructure Fundamentals

1.1 Evolution of Trading Systems

Until the 1990s, trading was dominated by open-outcry floors (NYSE, the CBOT and CME pits). NASDAQ was the exception, electronic from the start. Electronic adoption spread gradually through the 1990s and 2000s, and since then electronic matching and algorithmic trading have dominated. The move from floor to screen changed nearly every dimension of market microstructure: execution speed (minutes to microseconds), the transparency of order flow, the role of market makers, the mechanics of price discovery, and the level of transaction costs and spreads.

Once shares have been issued in the **primary market** (IPOs and follow-on offerings), they trade in the **secondary market**. In the US, secondary trading in equities takes place in three distinct kinds of venue, which we refer to as the second, third, and fourth markets.

Where US equities trade.

	Venue	Pre-trade transparency	Examples
Second market	Registered national securities exchanges	Lit: displayed quotes feed the NBBO	NYSE, Nasdaq, BATS, IEX
Third market	Over-the-counter (OTC) dealer networks	Dealer quotes; no central order book	OTC Markets Group, wholesale market makers
Fourth market	Dark pools and alternative trading systems (ATSs)	Dark: orders not displayed	Sigma X, UBS ATS, Liquidnet

1.2 The Second Market: Exchanges

The **second market** is the trading of listed equities on registered national securities exchanges. Exchanges run lit, continuous limit order books, publish their quotes to the consolidated feed, and are regulated by the SEC as self-regulatory organisations.

Exchange	Products	Key features
NYSE	Equities, ETFs	Hybrid market with designated market makers
Nasdaq	Equities, ETFs	Fully electronic, competing market makers
NYSE Arca	Equities, ETFs	All-electronic, price–time priority; leading ETF listing venue
BATS (Cboe BZX, BYX, EDGX, EDGA)	Equities, ETFs	All-electronic, price–time priority; launched by HFT firms in 2005 and acquired by Cboe in 2017; BYX uses inverted (taker–maker) pricing
IEX	Equities, ETFs	Speed bump and other anti-HFT features

1.3 The Third Market: Over-the-Counter (OTC) Equities

The **third market** is equity trading away from exchanges, through dealer networks. Dealers quote and trade as principal, and there is no central limit order book. It covers securities not listed on an exchange, which trade on OTC Markets Group, and also exchange-listed stocks executed off-exchange by wholesale market makers. The market is regulated by FINRA and is characterised by lower liquidity, wider spreads, and access to international companies.

OTC Markets Group sorts its securities into three tiers by the quality and timeliness of disclosure:

Tier	Requirements and constituents
OTCQX	Top tier: high financial standards, best-practice governance, current disclosure, and a third-party sponsor. Over 400 companies, including Roche, Publicis, and the Grayscale Bitcoin Trust (GBTC).
OTCQB	Middle tier: early-stage companies with current financial reporting and a \$0.01 minimum bid price. Over 900 companies.
Pink	Bottom tier: no minimum standards; includes bankrupt companies and those with limited disclosure.

1.4 The Fourth Market: Dark Pools and Alternative Trading Systems

The **fourth market** is trading through alternative trading systems (ATSs), which match buyers and sellers directly without a dealer intermediary. ATSs are regulated as broker-dealers under Regulation ATS, not as exchanges. Their dominant form is the **dark pool**: a private trading venue in which order information is hidden until after execution, so that institutional investors can trade large blocks without revealing their intentions.

Characteristics.
Orders are not displayed to the market, so large orders do not move prices before they execute. Executions are typically priced at the lit-exchange midpoint or better, and the venues are designed for block trading (>10,000 shares).

Ownership.
Broker-dealer owned: Goldman Sachs Sigma X, UBS ATS, Credit Suisse CrossFinder. *Exchange-owned*: NYSE, Nasdaq, and Cboe dark offerings. *Independent*: Liquidnet (institution-to-institution). IEX began as a dark pool but became a registered exchange in 2016, so it now belongs to the second market.

Regulation.
US: Regulation ATS and Form ATS-N. Europe: MiFID II volume caps of 4% per venue and 8% of total market volume.

Dark pools in numbers. Dark pools account for roughly 15–18% of US equity volume and 8–12% in Europe. They deliver average price improvement of 0.2–0.5 bps, fill rates of 15–35% for institutional orders, and a 25–40% reduction in market impact relative to lit markets.

Benefits	Concerns
Better execution prices	Reduced price discovery
Reduced information leakage	Market fragmentation
Lower transaction costs	Predatory trading risks
Confidential strategy execution	Regulatory opacity

1.5 Order Matching Mechanisms

When an incoming order crosses the book, the venue's matching algorithm decides which resting orders at the touched price get filled. Price priority is universal; the mechanisms differ in how they break ties within a price level.

Mechanism	Tie-break rule	Where used
Price–time	Earliest arrival at the price fills first	Most equity markets
Price–size	Larger orders at the price fill first	Some institutional markets
Pro rata	Fills allocated in proportion to resting size	Common in futures (CME SOFR)
Modified pro rata	A portion to the first order by time priority, the remainder pro rata	CME FX futures, Eurex

Pro rata worked example (CME SOFR futures). Resting bids at \$99.50: A = 1000, B = 500, C =

1500 contracts (total 3000). An incoming sell of 900 contracts is allocated

$$A : 900 \times \tfrac{1000}{3000} = 300, \qquad B : 900 \times \tfrac{500}{3000} = 150, \qquad C : 900 \times \tfrac{1500}{3000} = 450.$$

Implied pricing. Exchanges such as ICE and CME automatically create prices for calendar spreads and other combinations from the prices of the outright contracts. This increases liquidity, tightens bid–ask spreads, makes price discovery more efficient, and reduces arbitrage opportunities.

1.6 Order Types

Category	Order type	Behaviour
Basic	Limit	Buy or sell a specified quantity at a maximum (bid) or minimum (ask) price
	Market	Buy or sell a specified quantity at the current prevailing price
Time in force	Day	Expires at the end of the regular trading day
	GTC / GTT	Good till cancelled / good till time: requires manual cancellation to expire
	IOC	Immediate-or-cancel: executes as much as possible at the best price; the exchange cancels the rest
	FOK	Fill-or-kill: executes at the best price in its entirety or not at all
Display and routing	Iceberg	Partially displays its quantity; usually replenished up to a proportion when executed against
	Non-routable	Avoids routing to other exchanges (non-proactive)
	Pegged, hide-not-slide	Moves with the midpoint or the best bid/offer
	Discretionary	Displayed at one price but may execute at a less favourable (hidden) price
Auction	MOC	Market on close: participates in the closing auction as a market order
	LOC	Limit on close: participates in the closing auction as a limit order

2 Stylized Facts of Equity Markets

Five empirical regularities of equity returns recur across markets and sample periods. Below, r_t denotes the log return.

Absence of autocorrelations.
Returns show little to no serial correlation at daily or higher frequencies, consistent with market efficiency, although intraday patterns may exist:

$$\mathrm{Corr}(r_t, r_{t-k}) \approx 0 \quad \text{for } k > 0.$$

Fat tails (leptokurtosis).
Return distributions have heavier tails than the normal, so extreme moves are more likely. Excess kurtosis is typically 3–10 (against 0 for the normal), the tail index is often between 3 and 4 (finite variance, with the fourth moment borderline infinite), and a Student's t distribution usually fits better than the normal.

Volatility clustering (Taylor effect).
High-volatility periods tend to follow high-volatility periods, which shows up as strong correlation in absolute and squared returns:

$$\mathrm{Corr}(|r_t|, |r_{t-k}|) > 0, \qquad \mathrm{Corr}(r_t^2, r_{t-k}^2) > 0.$$

The autocorrelation of absolute returns decays slowly, often remaining significant for dozens of lags.

Aggregational Gaussianity.
Daily returns are non-normal, but longer-horizon returns (monthly, quarterly) approach normality through central limit effects. Risk models can therefore use normal distributions at long horizons while needing fat-tailed distributions for daily risk.

Leverage effect.
Returns are negatively correlated with subsequent changes in volatility: a fall in the stock price raises future volatility more than an equivalent rise. Explanations include the increase in financial leverage as equity values decline, behavioural factors (panic selling, margin calls), and information asymmetry.